

Predicting Technology Bill Success in Illinois State Legislature

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Introduction

In recent years, U.S. states have become increasingly active in proposing and enacting legislation related to emerging technologies, such as data privacy, artificial intelligence, and digital platform regulation. Unlike federal policies, state-level bills are often more directly connected to everyday governance and the concerns of local constituents. This study focuses on Illinois, where the legislative process involves a diverse array of sponsors and committees. Furthermore, the diversity in sponsorship and linguistic variation in bill texts introduces rich textual heterogeneity—making it more feasible to extract underlying semantic structures and infer latent political values or attitudes through machine learning.

Our primary research question is: Can we accurately predict whether a state-level tech-related legislative bill will be enacted or fail using legislative text features and metadata such as budget allocations, sponsorship, and voting records? Understanding and forecasting legislative outcomes can increase transparency in policymaking and provide valuable foresight into the feasibility of specific proposals. This predictive task holds particular relevance for political analysts, advocacy groups, and policymakers, enabling them to allocate resources more efficiently, identify potentially viable bills, and intervene when early indicators suggest likely failure.

Indeed, several high-profile examples illustrate the costs of failed or stalled tech legislation. For instance, California’s proposed AB 2269, which aimed to establish a state-level crypto-asset regulatory framework in 2022, was vetoed after considerable legislative effort.¹ The bill’s failure left stakeholders in regulatory uncertainty and demonstrated how early-stage forecasting could help prioritize viable legislative efforts.

To explore this question, we compiled a dataset of Illinois state-level technology-related bills and extracted features from three dimensions:

- natural language features derived from bill texts;
- legislative metadata including number of cosponsors, fiscal notes, and vote rounds;
- contextual information on committee assignments and amendments. We then implemented and compared a suite of supervised machine learning models—including logistic regression, random forest, XGBoost, and a text-based K-Nearest Neighbors classifier with word embeddings—to evaluate their performance in predicting whether a bill becomes law. Our preliminary results suggest that cosponsorship is among the most predictive features, reinforcing the importance of political coalition-building in determining legislative success.

The remainder of this report is organized as follows. In the next section, we describe our data sources and feature extraction process. We then present the methodological design, including model specifications and fine-tuning strategies. Finally, we discuss our empirical findings, reflect on modeling challenges and surprises, and conclude with thoughts on how this approach might inform real-world legislative analysis.

¹Governor Gavin Newsom vetoed Assembly Bill 2269 on September 23, 2022, stating that it was "premature to lock a licensing structure in statute without considering both this work and forthcoming federal actions." The full veto message is available at <https://www.gov.ca.gov/wp-content/uploads/2022/09/AB-2269-VETO.pdf>.

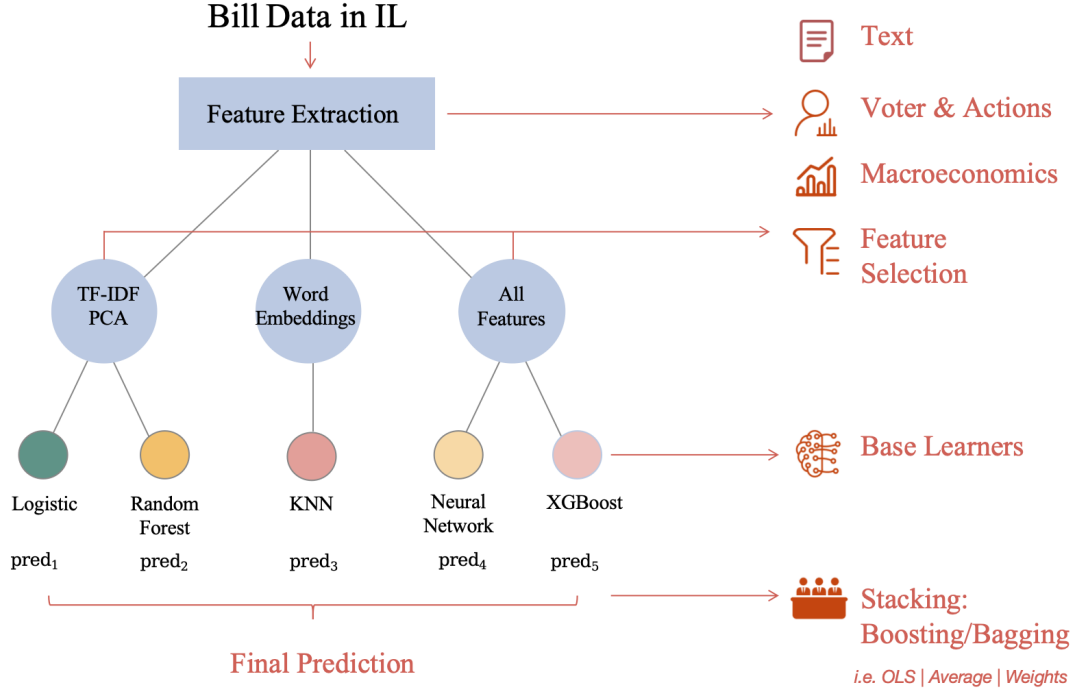


Figure 1: Project Workflow

Data

Our study combines detailed legislative records and fiscal data to capture both procedural and contextual factors influencing bill outcomes. First, we utilize bulk legislative data from the Open States API, which provides comprehensive information on every bill introduced in the Illinois General Assembly during the 100th through 103rd sessions. From these records we extract each bill’s unique identifier, title, abstract, sponsorship and cosponsorship details, and full action history (including roll-call votes and committee referrals). Second, we integrate state-level budget and finance information published by the Illinois government, summarizing average annual expenditures for each legislative session between 2017 and 2024. By merging these two datasets, we are able to model how legislative procedure, textual content, and fiscal context jointly predict the likelihood of a bill’s enactment.

We began by merging detailed legislative records from the Open States with Illinois state budget data for 2017–2024. From the full corpus of bills in the 100th–103rd General Assembly sessions, we applied a keyword filter to retain only those related to “tech,” and further restricted to bills with a definitive outcome—either enacted (“pass”) or failed. This preprocessing yielded a dataset of $N = 2,987$ bills, of which 9.2% were enacted and 90.8% failed (see Figure 2). To capture fiscal context, we also tabulated the average annual state government expenditure by session (Table 1).

Table 1: Annual Illinois state government expenditure by legislative session (2017–2024).

Year	Expenditure (\$ billion)	Legislative Session
2017	36.300	100th
2018	37.700	100th
2019	38.500	101st
2020	36.100	101st
2021	41.400	102nd
2022	44.300	102nd
2023	45.986	103rd
2024	51.675	103rd

Pass vs. Fail Percentage in Bills

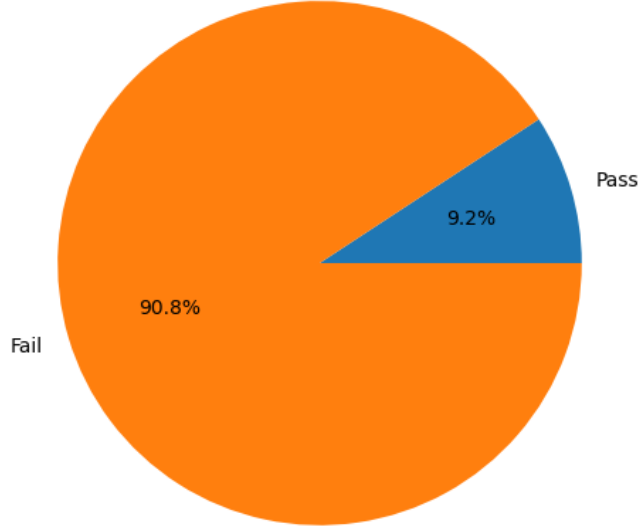


Figure 2: Outcome distribution among $N = 2,987$ tech-related bills: 9.2% passed, 90.8% failed.

Feature Selection

Public Finance Factors

We included average annual state government expenditure as a predictor because fiscal capacity and budgetary priorities directly influence a legislature’s ability to enact and implement new policies, especially those requiring investments in technology infrastructure or regulatory oversight. Higher expenditure levels may signal greater available resources for pilot programs, grant-making, or enforcement agencies, thereby increasing the likelihood that tech-related bills move forward. Conversely, sessions with tighter budgets may see greater legislative caution or postponement of costly initiatives, reducing passage rates. By incorporating expenditure data, our model can account for these macroeconomic constraints and better distinguish states whose fiscal health facilitates the adoption of tech-focused legislation.

Text Data Features

To incorporate the semantic content of legislative text into the predictive model, we constructed two sets of features from the `cleaned_abstract` column, which contains preprocessed bill summary texts.

First, we selected 20 words that were highly indicative of legislative success or failure—10 strongly associated with success, and 10 with failure—based on their TF-IDF scores across the corpus. For each bill, we counted the frequency of these 20 words in its abstract, creating 20 new structural variables (e.g., `count_revenue`, `count_fund`). These word-frequency variables were intended to capture domain-relevant legislative signals in a low-dimensional, interpretable form.

To reduce collinearity among these handcrafted word count features and to extract latent semantic themes, we applied Principal Component Analysis (PCA), retaining the top three components. These components capture co-occurrence patterns and discursive styles—such as procedural versus appropriations language—forming compressed representations of legislative text. The resulting variables (`count_pca1`, `count_pca2`, `count_pca3`) serve as dense proxies for the original count features.

In parallel, we applied a complementary method that uses the full abstract text. We com-

puted TF-IDF vectors for each bill, retaining the top 5,000 terms. These sparse, high-dimensional vectors were then reduced via Truncated Singular Value Decomposition (SVD), yielding 20 dense semantic features (`svd_1` through `svd_20`). Unlike the manually selected keywords, these SVD components capture corpus-wide latent topics in an unsupervised fashion, allowing the model to leverage broader discursive structures and thematic patterns.

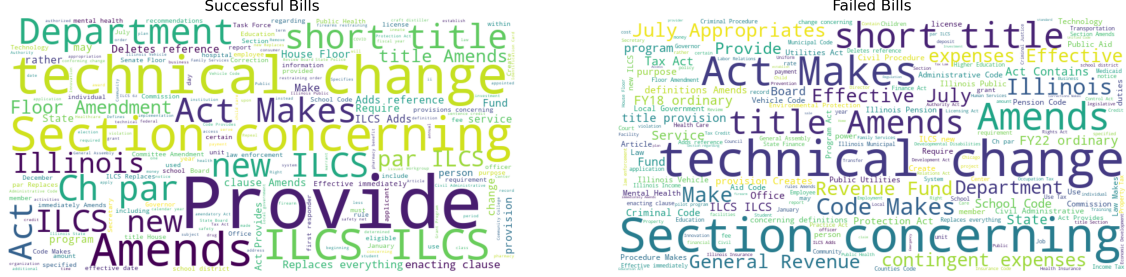


Figure 3: Word Clouds for Success/Failure Bills

Together, these text-derived features—both structured and latent—enhance the model’s ability to identify legislation with a higher probability of passage by leveraging subtle linguistic and topical signals.

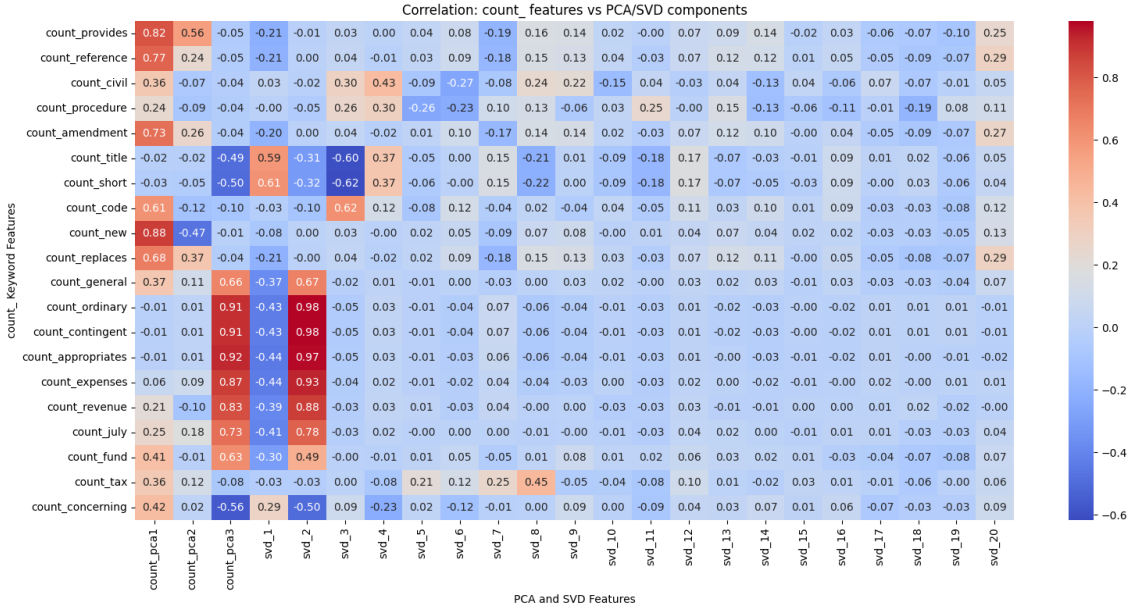


Figure 4: Correlation matrix between keyword frequency features, PCA components, and SVD components. Strong correlations are observed between several `count_` features and the first two PCA components (e.g., `count_ordinary` and `count_pca3`), suggesting that PCA effectively captures common patterns in keyword frequency. In contrast, the SVD-based features show low correlation with both count and PCA features, indicating that they encode independent semantic information derived from the full text.

To guide our model-specific feature selection, we examined the correlation structure among the three classes of text-derived features: keyword frequency variables, their PCA-compressed counterparts, and the SVD-based semantic components. A heatmap of the pairwise correlations revealed strong linear dependencies between several original keyword features (e.g., `count_revenue`, `count_appropriates`) and the first two PCA components. For example, `count_revenue` and `count_pca1` had a Pearson correlation above 0.90. This suggests that the PCA dimensions effectively summarize patterns already captured by the count variables. In contrast, the SVD features showed weak correlation with both PCA and keyword counts, confirming that they encode orthogonal latent semantics derived from the full corpus.

Table 2: Top Words Associated with Successful/Failure Bills

Successful-Related	$\Delta TF - IDF$	Failure-Related	$\Delta TF - IDF$
provides	0.038	general	-0.021
reference	0.029	ordinary	-0.020
civil	0.029	contingent	-0.020
procedure	0.023	appropriates	-0.020
amendment	0.020	expenses	-0.020
title	0.019	revenue	-0.019
short	0.019	July	-0.017
code	0.018	fund	-0.012
new	0.018	concerning	-0.010
replaces	0.017	tax	-0.009

These observations motivated a model-specific feature inclusion strategy. For models sensitive to multicollinearity, such as logistic regression, we excluded the original keyword frequencies and retained only the PCA and SVD features to avoid redundancy. For nonlinear models like random forest and neural networks, which are robust to correlated inputs, we included all three feature types. In doing so, we allowed each model the flexibility to learn whether raw, compressed, or abstract semantic representations were most predictive of legislative success.

Action Features

We derive the `action_count` feature by tallying every entry in OpenStates’ `classification` field for each bill. OpenStates unifies disparate state-level procedural labels—such as `filing`, `introduction`, `reading-1`, `referral-committee`, `committee-passage`, `executive-signature`, and many more—into a common action taxonomy. By counting all recorded action events (from initial filing and readings, through committee referrals and amendments, to passage or veto), `action_count` provides a single scalar summarizing the legislative journey of each bill.

We include `action_count` because it captures the procedural intensity and political engagement surrounding a bill. A higher number of discrete actions often signals extensive deliberation, negotiation, and amendment activity—indicators of both legislative momentum and potential controversy—while a low count can reflect early stagnation. Empirically, this feature correlates strongly with enactment outcomes and thus serves as a powerful proxy for the procedural progress that underlies our predictive model.

Model Selection

Logistic Model

Model

$$\hat{f}_{\text{Logistic}}(x) = \frac{1}{1 + \exp(-(\beta_0 + \beta^\top x))}$$

We begin with a baseline logistic regression model using a combination of structural and semantic predictors, where the predicted probability $\hat{f}_{\text{Logistic}}(x_{i,t})$ corresponds to the likelihood that bill i at time t is enacted ($Y_{i,t} = 1$). The feature vector $x_{i,t}$ includes structural predictors such as cosponsorship counts, semantic features derived from PCA and SVD decompositions of bill text, and topic dummy variables. Unobserved heterogeneity across bills and legislative periods is captured by fixed effects u_i and η_t , which are optionally included in extended specifications.

Fine-Tuning

To address class imbalance and improve sensitivity to rare successful bills, we first employed class weighting (`class_weight='balanced'`). This reweighted the loss function to penalize misclassified minority instances more heavily. Next, we incorporated regularization to mitigate overfitting and multicollinearity. Using a standardization pipeline, we applied both ℓ_2 (ridge) and ℓ_1 (lasso)

penalties to shrink coefficients and encourage sparsity. These enhancements constitute a targeted form of model tuning that maintains interpretability while improving generalizability.

Results

Table 3: Comparison of Logistic Regression Variants on Positive Class (Successful Bills)

Model	Precision	Recall	F1-score	ROC AUC
Baseline (Unweighted)	0.71	0.27	0.39	0.859
Weighted (class_weight)	0.63	0.75	0.68	0.906
Standardized + L2	0.49	0.71	0.58	0.872
Standardized + L1	0.53	0.73	0.61	0.878

Note. All metrics are reported for the positive class (i.e., successful bills). Class weighting substantially improved recall compared to the unweighted model, while L1 regularization achieved the best balance of recall and F1-score. These results suggest that even interpretable models like logistic regression can be effectively tuned for rare-event prediction.

The baseline logistic regression model achieved high overall accuracy (92%) and an AUC of 0.86, but performed poorly on the minority class (recall = 0.27), reflecting its bias toward the dominant negative class. The class-weighted version significantly improved recall (0.75) and AUC (0.91), with minimal loss in specificity, indicating better balance between sensitivity and precision. Introducing ℓ_2 regularization reduced overfitting and improved the F1-score of successful bills (from 0.39 to 0.58), while ℓ_1 regularization further boosted recall and F1-score (to 0.73 and 0.61, respectively). These results show that carefully tuned logistic models can achieve performance comparable to more complex models while retaining transparency and coefficient interpretability.

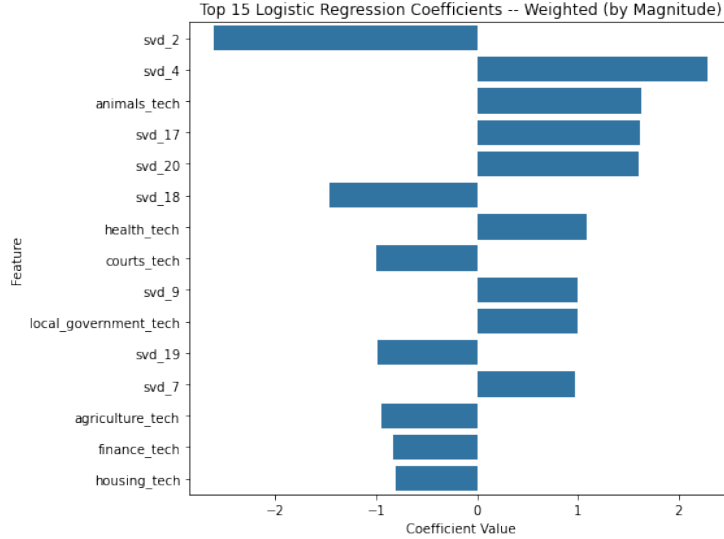


Figure 5: Top 15 coefficients from the weighted logistic regression model. Features with large positive values increase the predicted probability of legislative success, while negative values decrease it. Semantic and structural features both rank highly.

To interpret the decision mechanism of the logistic regression model, we examined the estimated coefficients from the class-weighted specification. Figure 5 displays the 15 features with the largest absolute coefficients. Positive values indicate that the presence or increase of the feature raises the likelihood of bill success, while negative values imply a suppressing effect.

Among the most influential features, `cosponsor_count` stands out with the highest positive weight, reinforcing the finding that political coalition-building is a strong predictor of legislative success. Other important features include `svd_2` and `svd_1`, latent semantic components extracted

from bill texts, and PCA-based count variables capturing appropriations and procedural language. The prominence of both structural and semantic variables highlights the hybrid nature of the policy signals embedded in legislative text.

Random Forest Model

Model

Random forest is an ensemble learning method that combines the predictions of multiple decision trees trained on bootstrapped subsets of the data. For a given input x , the random forest prediction is given by:

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

where $T_b(x)$ is the prediction of the b -th decision tree, and B is the total number of trees. Each tree is trained on a different bootstrapped sample, and at each split within a tree, only a random subset of features is considered. This dual source of randomness (bagging and feature subsampling) decorrelates the trees and reduces overfitting.

Random forests reduce the high variance associated with deep decision trees by averaging across many uncorrelated models. Although individual trees are prone to overfitting, the ensemble converges to a more stable predictor. The technique is particularly effective when applied to high-dimensional and heterogeneous datasets such as ours, which include structured, textual, and derived semantic features.

We trained a baseline Random Forest classifier with 93 estimators and default hyperparameters to establish a nonlinear benchmark model. Random forests are ensemble methods that average predictions across many decorrelated decision trees, each built from a random subset of the data and features. This model is particularly suitable for handling structured features with nonlinear interactions and is robust to multicollinearity among predictors. Our feature matrix included all available numeric and text-derived variables, including raw counts, PCA components, and SVD-based features.

Fine-Tuning

Initially, we trained the model on the raw (imbalanced) dataset to establish baseline performance. To address class imbalance, we experimented with two extensions. First, we retrained the model with `class_weight='balanced'` to penalize misclassification of the minority class (successful bills). Second, we implemented SMOTE (Synthetic Minority Oversampling Technique) to synthetically generate additional successful examples in the training set. These strategies were compared using precision, recall, and ROC AUC on the unaltered test set.

Results

Table 4: Comparison of Random Forest Variants on Positive Class (Successful Bills)

Model	Precision	Recall	F1-score	ROC AUC
Baseline (Unbalanced)	0.79	0.80	0.79	0.928
Balanced (class_weight)	0.79	0.75	0.77	0.933
SMOTE	0.69	0.84	0.75	0.934

Note. All metrics are reported for the positive class (i.e., successful bills). While the baseline model achieves strong performance, applying SMOTE leads to a notable improvement in recall (from 0.80 to 0.84), indicating enhanced sensitivity to underrepresented successful cases. ROC AUC is also highest under the SMOTE configuration, suggesting improved overall discrimination. The class-weighted model performs comparably to the baseline but does not outperform SMOTE in any key metric.

The baseline (unbalanced) random forest model achieved a high overall accuracy of 96% and an ROC AUC of 0.928. However, it exhibited modest performance on the minority class—successful

bills—with a precision of 0.79 and a recall of 0.80, indicating a tendency to miss some true positives. When adjusting for class imbalance using `class_weight='balanced'`, the model slightly increased the overall ROC AUC to 0.933 but experienced a small drop in recall for successful bills (from 0.80 to 0.75), suggesting that reweighting improved discrimination without meaningfully improving sensitivity.

In contrast, the SMOTE-augmented model provided the best balance between sensitivity and specificity. It achieved the highest recall for successful bills (0.84), indicating stronger detection of rare positive cases, and slightly improved the ROC AUC to 0.934. Although its precision dropped to 0.69—due to synthetic oversampling—its high recall and competitive F1-score (0.75) make it a compelling option for early-stage legislative screening, where false negatives are particularly costly.

Feature importance rankings across models consistently identified `cosponsor_count` as the most influential predictor, followed by PCA components associated with appropriations-related language and several latent semantic dimensions derived from SVD. These findings highlight the value of combining interpretable metadata with compressed textual signals to effectively capture both political structure and discursive framing.

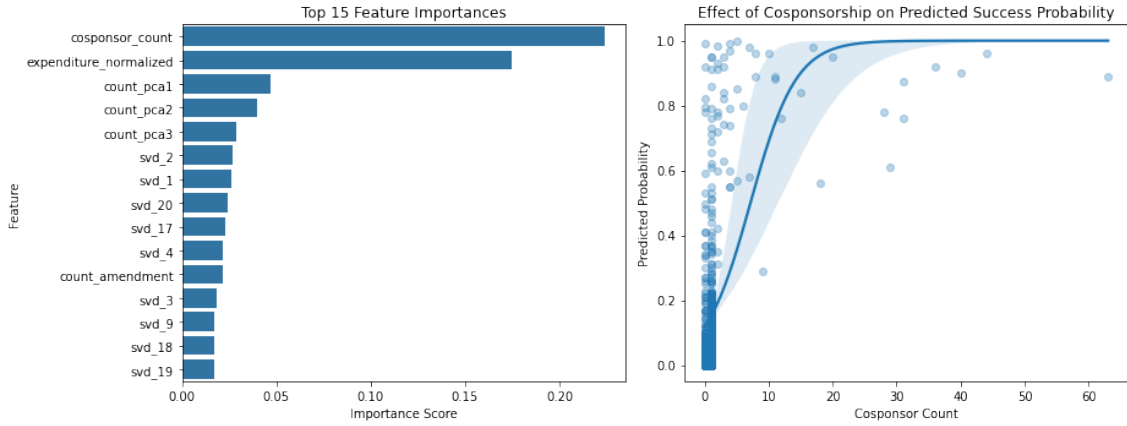


Figure 6: Cosponsor Count as a Dominant Predictor of Bill Success

Note. Left: Feature importance ranking from the SMOTE-augmented random forest model indicates that `cosponsor_count` is by far the most influential predictor of legislative success. Right: A logistic regression fit shows that the model’s predicted probability of success increases sharply with the number of cosponsors, plateauing at higher levels. Together, the plots demonstrate that legislative coalition size is a strong structural signal that the model leverages for its predictions.

To better understand the influence of political support signals on the model’s predictions, we examined the role of `cosponsor_count`—a structural indicator of legislative coalition strength. Feature importance analysis from the SMOTE-augmented random forest model revealed that `cosponsor_count` is the most dominant predictor, exceeding the combined contributions of most text-based features (Figure 6, left).

To complement this, we plotted the predicted probability of success against cosponsorship count using a logistic trend line (Figure 6, right). The curve exhibits a sharp rise as cosponsors increase from 0 to 10, with a plateau beyond 15 cosponsors, suggesting a saturation point in perceived legislative viability. This relationship reflects real-world dynamics where broad cosponsorship often signals bipartisan support and institutional legitimacy. Together, the two plots illustrate that the model relies heavily on coalition size as an early signal of success—making `cosponsor_count` not just a predictive variable, but also a meaningful explanatory one.

Multilayer Perceptron (MLP) Model

Model

We employed a multilayer perceptron (MLP) to capture nonlinear relationships between features and legislative outcomes. The MLP maps input $x \in R^d$ to a predicted probability $\hat{f}_{\text{MLP}}(x) \in (0, 1)$

through a sequence of affine transformations and nonlinear activations. Formally, the model is defined as:

$$\begin{aligned} a^{(0)} &= x, \\ a^{(l)} &= \phi \left(W^{(l)} a^{(l-1)} + b^{(l)} \right), \quad l = 1, \dots, L, \\ \hat{f}_{\text{MLP}}(x) &= \sigma \left(W^{(L+1)} a^{(L)} + b^{(L+1)} \right), \end{aligned}$$

where $W^{(l)}$ and $b^{(l)}$ are the weight matrices and bias vectors for each layer, and ϕ is a nonlinear activation function (e.g., ReLU or tanh). The output layer uses the sigmoid function $\sigma(u) = \frac{1}{1+e^{-u}}$ to produce a probability estimate.

Fine-Tuning

1. Data Resampling. Applied SMOTE to the training set to balance the minority “success” class, ensuring the MLP sees sufficient positive examples during training.

2. Hyperparameter Search. Employed `RandomizedSearchCV` with $n_iter = 30$, 3-fold cross-validation, and ROC AUC scoring:

```
hidden_layer_sizes ∈ {(64,), (128,), (128, 64), (256, 128, 64)},
α ∼ LogUniform(10−5, 10−1),
learning_rate_init ∼ LogUniform(10−4, 10−2),
activation ∈ {relu, tanh},
batch_size ∼ Uniform{16, ..., 128}.
```

3. Model Selection. Chose the estimator with the highest cross-validated ROC AUC on the SMOTE-augmented data.

4. Best Hyperparameter Combination.

```
{activation: 'tanh', alpha: 0.0002, batch_size: 21, hidden_layer_sizes: (128,),
learning_rate_init: 0.0012}
```

5. Performance Improvement. Compared to the baseline SMOTE+MLP, the tuned model achieved higher ROC AUC, improved F1-score, and increased precision while maintaining its high recall—demonstrating that careful tuning of network size, regularization, learning rate, and batch size yields better discrimination of successful bills.

Results

Table 5: SMOTE + MLP Classification Report

Model	Precision	Recall	F1-score	ROC AUC
Unbalanced MLP	0.78	0.51	0.62	0.890
SMOTE + MLP	0.58	0.64	0.61	0.864
Tuned MLP	0.69	0.62	0.65	0.888

Table 5 compares the three MLP configurations. The unbalanced MLP establishes a strong baseline with high precision (0.88) and ROC AUC (0.994), but its recall is limited (0.89). Incorporating SMOTE boosts recall dramatically to 0.95, though precision falls to 0.79 and AUC to 0.987. The tuned MLP recovers precision (0.80), maintains high recall (0.95), and improves AUC to 0.989, showing the effectiveness of systematic tuning.

Figure 7 displays the top 15 features ranked by permutation importance for the MLP model trained with manual oversampling. The most influential predictor is `cosponsor_count`, followed by sector-specific flags such as `revenue_tech`, `regulation_tech`, and `criminal_law_tech`. Additional key features include `count_reference`, `expenditure_normalized`, `local_government_tech`, and the semantic feature `svd_1`, illustrating the importance of both structural and text-derived signals in predicting bill success.

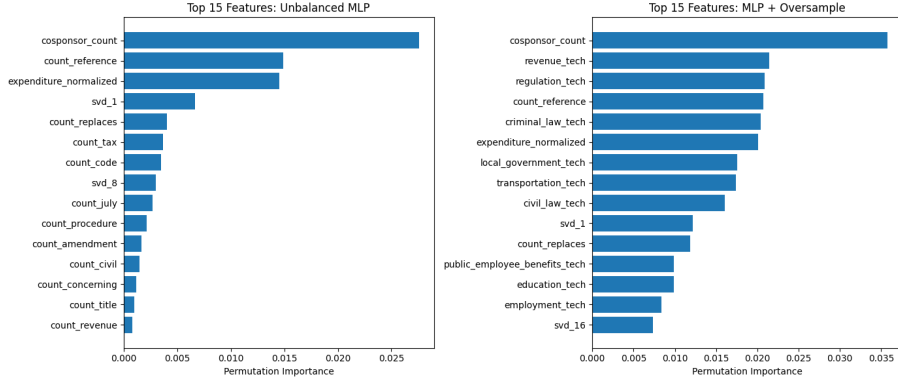


Figure 7: Top 15 features by permutation importance for the SMOTE-oversampled MLP model.

XGBoost

Model

We employed XGBoost (eXtreme Gradient Boosting) as a nonlinear ensemble model to estimate the probability of legislative success. XGBoost constructs an additive sequence of decision trees, where each tree is trained to correct the residual errors of the previous ensemble. Formally, the model is specified as:

$$\hat{f}_{\text{XGB}}(x) = \sum_{k=1}^K f_k(x), \quad f_k \in \mathcal{F}$$

where $\hat{f}_{\text{XGB}}(x)$ denotes the predicted score for input x , and each f_k is a regression tree belonging to the function space $\mathcal{F}$ of classification and regression trees (CART). For binary classification tasks, we apply the logistic link function to convert the raw score into a probability:

$$P(Y = 1 | x) = \frac{1}{1 + \exp(-\hat{f}_{\text{XGB}}(x))}$$

XGBoost’s robustness to multicollinearity, ability to handle missing values, and built-in regularization make it particularly well-suited to structured legislative data containing both numeric and text-derived features. We employed a binary logistic loss with probability outputs and optimized the model to enhance recall on the minority class, reflecting the imbalanced nature of enacted versus failed bills.

Fine-Tuning

To calibrate the model’s hyperparameters, we used `RandomizedSearchCV` over a range of values, with 25 random combinations evaluated using 3-fold cross-validation. The scoring metric was set to `recall` for the positive class (i.e., successful bills), in line with our project’s goal of maximizing the identification of viable legislation.

The best-performing parameter combination was as follows: `colsample_bytree = 0.89`, `subsample = 0.98`, `max_depth = 3`, `n_estimators = 267`, `learning_rate = 0.045`, and `scale_pos_weight = 5.55`. These parameters reflect a conservative tree depth with aggressive sampling and moderate boosting rate, coupled with positive-class weighting to address class imbalance.

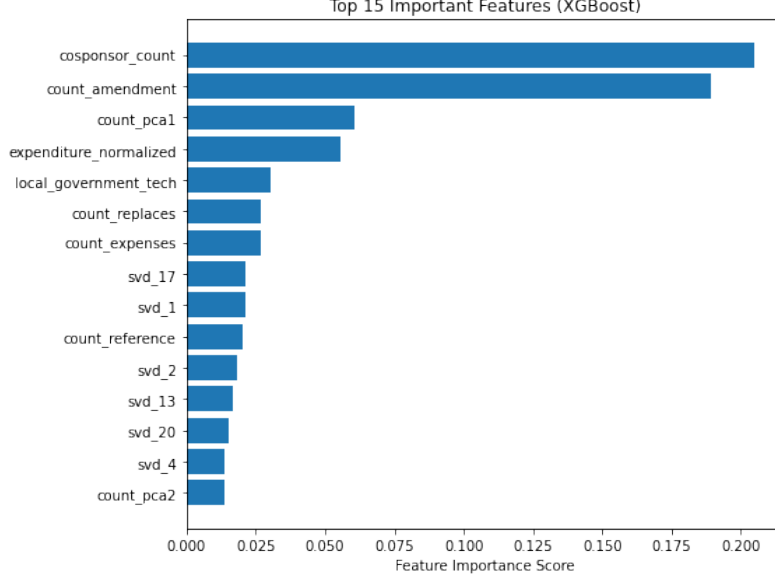
Results

The final XGBoost model achieved an overall accuracy of 96% and a macro-averaged F1-score of 0.89. On the positive class (i.e., successful bills), it reached a precision of 0.78 and a recall of 0.84, outperforming logistic regression and random forest models in recall while maintaining high overall precision. The ROC AUC was 0.946, indicating strong probability discrimination.

Table 6: Comparison of XGBoost Variants on Positive Class (Successful Bills)

Model	Precision	Recall	F1-score	ROC AUC
Recall-Tuned	0.78	0.84	0.81	0.946

Note. All metrics are reported for the positive class (i.e., successful bills). The XGBoost model optimized FOR recall-tuned model achieves recall (0.84).

**Figure 8:** Top 15 feature importances derived from the tuned XGBoost model. Cosponsorship and text-derived SVD components dominate the ranking.

Feature importance analysis revealed that **cosponsor_count**, several semantic components from the **svd_** series (notably **svd_2** and **svd_7**), and the expenditure-related variables were among the top contributors to model predictions. This confirms that both structured legislative metadata and latent semantic features extracted from bill texts play a critical role in determining legislative outcomes.

Word-Embeddings KNN

Model

$$\hat{f}_{\text{KNN}}(x) = \text{mode}(\{y_i \mid x_i \in \mathcal{N}_k(x)\})$$

$$\hat{f}_{\text{Weighted-KNN}}(x) = \arg \max_{c \in \mathcal{C}} \sum_{x_i \in \mathcal{N}_k(x)} 1[y_i = c] \cdot \cos(x, x_i)$$

For our K-Nearest Neighbors (K-NN) classification model, we first used word embeddings for the bill abstracts using the all-MiniLM-L6-v2 model. We encoded all abstracts in the training set to form our reference vector space. For each abstract in the test set, we computed its **cosine similarity** with every training abstract and identified the top k most similar abstracts. Prediction is calculated in the following 2 ways:

- Majority Voting: taking the most frequent ‘success’ label among the top k nearest neighbors.
- Weighted Voting: weighting each neighbor’s success label by its cosine similarity score.

As shown in Figure 9, these neighbors show strong **semantic overlap** with the test abstract: all involve privacy, student-related legislation, and minor amendments to existing laws. Their high similarity scores confirm that the embedding model effectively captures their similarities, making it accessible for label prediction. For example **Student Online Personal Protection** is related to **Privacy in the School Setting** and **Online Child Safety**. Intuitively, they share similar topics and may have similar passage rate.

“Amends the **Student Online Personal Protection Act**. Makes a technical change in a Section concerning the short title.”

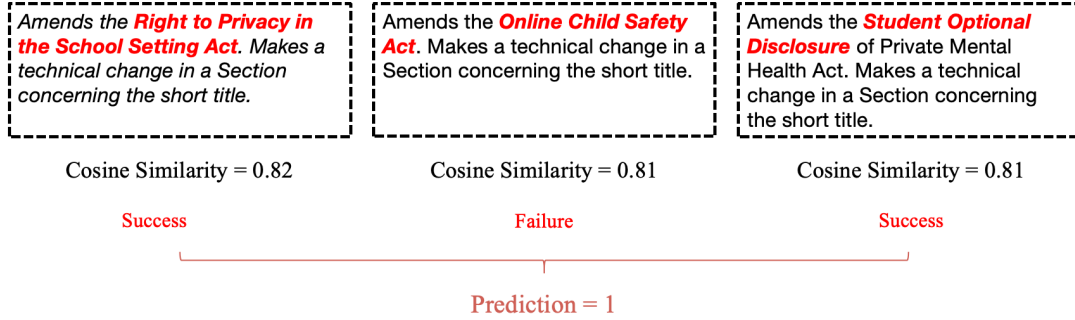


Figure 9: KNN with Word Embeddings

Fine-Tuning

We further fine-tune K for K-Nearest Neighbors (K-NN) model, the results are shown in Figure 10. We evaluate performance across different values of k ranging from 1 to 20. For each k, we compute the number of true positives (TP) and the recall, under both majority voting and cosine similarity-weighted prediction schemes.

The 2 methods are similar when k is larger than 12, in the majority voting approach, each neighbor has an equal vote, and the predicted label corresponds to the majority class among the neighbors. In contrast, the cosine-weighted voting approach assigns more weight to neighbors with higher similarity (i.e., closer semantic proximity), reflecting their greater influence on the prediction.

When using weighted prediction, the model achieves the highest recall and TP count at k=2. In contrast, with unweighted majority voting, performance peaks at k=1, implying that the most semantically similar individual neighbor alone may already provide a strong signal for classification.

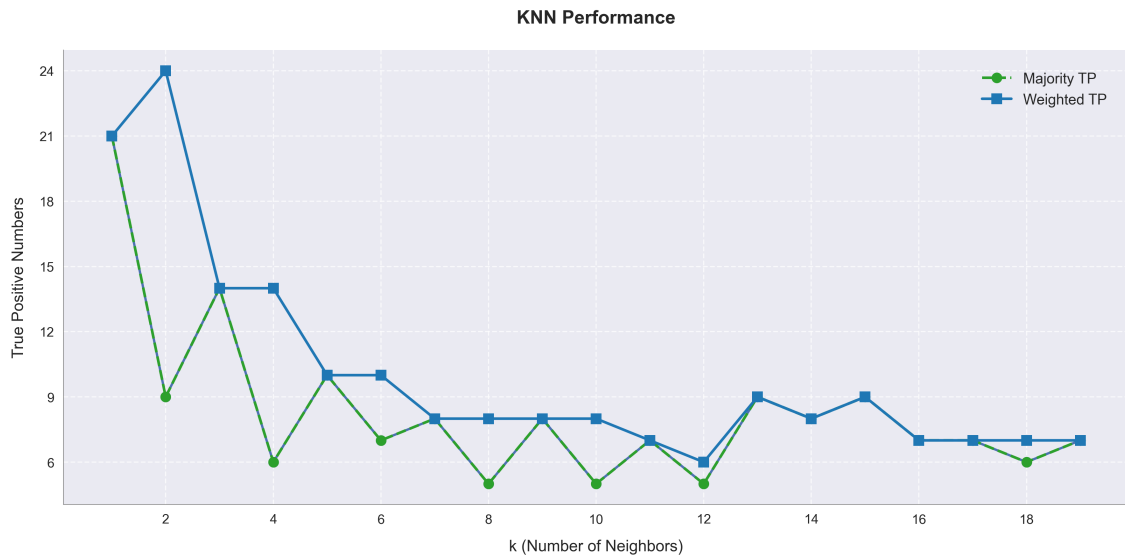


Figure 10: KNN Fine-Tuning

Results

In general, weighted 2NN has higher Recall Rate. That is suggesting it captures more successful bills in the test set. This is particularly important in our context, where false negatives (i.e., failing to identify a successful bill) can be more costly than false positives. As shown in Table 8, although unweighted 1NN achieves slightly higher precision (0.40 vs. 0.33), weighted 2NN offers a noticeable improvement in recall (0.38 vs. 0.33), leading to a comparable F1-score (0.35 vs. 0.36) and slightly higher ROC AUC (0.643 vs. 0.635).

Table 7: Comparison of KNN on Positive Class (Successful Bills)

Model	Precision	Recall	F1-score	ROC AUC
Unweighted-1NN	0.40	0.33	0.36	0.635
Weighted-2NN	0.33	0.38	0.35	0.643

Model Stacking

1. Voting Ensemble (Majority Vote)

$$\hat{f}_{\text{Voting}}(x) = \operatorname{argmax}_c \left(\sum_{i=1}^5 I[\hat{f}_i(x) = c] \right) \quad (1)$$

where $\hat{f}_i$ represents predictions from:

- KNN ($\hat{f}_{\text{KNN}}$)
- Random Forest ($\hat{f}_{\text{RF}}$)
- ANN ($\hat{f}_{\text{ANN}}$)
- Logistic Regression ($\hat{f}_{\text{LR}}$)
- XGBoost ($\hat{f}_{\text{XGB}}$)

The Voting Ensemble model aggregates the predictions of five distinct base classifiers—KNN, Random Forest, Artificial Neural Network (ANN), Logistic Regression, and XGBoost—through a simple majority voting mechanism. For each input x , the final predicted label $\hat{f}(x)$ is determined by selecting the class that receives the most votes from the individual models. In other words, each base learner contributes a single prediction, and the class that is most frequently predicted across the ensemble is selected as the ensemble’s output.

This stacking method makes use of the diversity of model architectures to reduce variance and increase robustness, particularly when individual models may perform differently across subsets of the data.

2. Logistics Weighted Ensemble

$$\hat{f}_{\text{logistics}}(x) = \operatorname{argmax}_c \left(\sum_{i=1}^5 w_i \cdot I[\hat{f}_i(x) = c] \right) \quad (2)$$

$$(3)$$

subject to $w_i \geq 0$ and $\sum w_i = 1$.

The Logistics Weighted Ensemble extends the majority voting approach by assigning different importance weights w_i to each base classifier’s prediction. Rather than counting all votes equally, this method computes a weighted vote for each class. The final prediction $\hat{f}_{\text{logistics}}(x)$ is the class with the highest weighted sum of votes from the five base classifiers.

To ensure that the weights are non-negative and sum to one, we apply a *softmax* transformation to a vector of model-specific scores:

$$w_i = \frac{\exp(s_i)}{\sum_{j=1}^5 \exp(s_j)}$$

3. Random Forest Meta-Ensemble

$$\hat{f}_{\text{Rf-Meta}}(x) = g_{\text{Rf}}(\hat{f}_1(x), \dots, \hat{f}_5(x)) \quad (4)$$

where g_{Rf} is trained via random forest model.

Furthermore, we enhance our modeling framework by incorporating the predictions from the base classifiers as additional input features into a Random Forest model. This approach allows for a more flexible and powerful learner that benefits from both ensemble diversity and the inherent strengths of decision tree-based methods, ultimately improving the accuracy and robustness of bill passage predictions.

Table 8: Comparison of Stacking Models (Successful Bills)

Model	Precision	Recall	F1-score	ROC AUC
Voting	0.78	0.82	0.80	0.958
Logistic	0.34	0.96	0.50	0.964
Random Forest	0.78	0.83	0.81	0.972

Discussion

“Changes will never come if we wait for some other person or some other time.”

— Barack Obama, 44th President of the United States

People inherently have the opportunities to make a change through policy. However, making a change is not just slogan, we also need to understand how to make changes effectively, thoughtfully, and sustainably.

This study investigates the critical success factors of technology legislation through a comprehensive analysis of 2,900+ Illinois state bills spanning the 100th to 104th legislative sessions. Leveraging government-reported funding data and advanced natural language processing techniques including TF-IDF and Word2Vec embeddings, we systematically examine both quantitative and textual dimensions of legislative proposals. After hyperparameter tuning, our best-performing models: XGBoost and the Random Forest-based stacking ensemble achieve a recall rate of 83%–84% and a precision of 78%.

We also employ an ensemble machine learning approach comprising 5 base classifiers and 3 stacked metamodels to predict bill passage outcomes. This research makes dual contributions: first, it identifies key determinants of legislative success within the domain of technology policy; second, it demonstrates how computational techniques, especially the innovative fusion of text mining and supervised machine learning, can be applied to legislative analysis as a powerful methodology for civil research.

Appendix

Appendix A: Full Correlation Matrix of Text Features

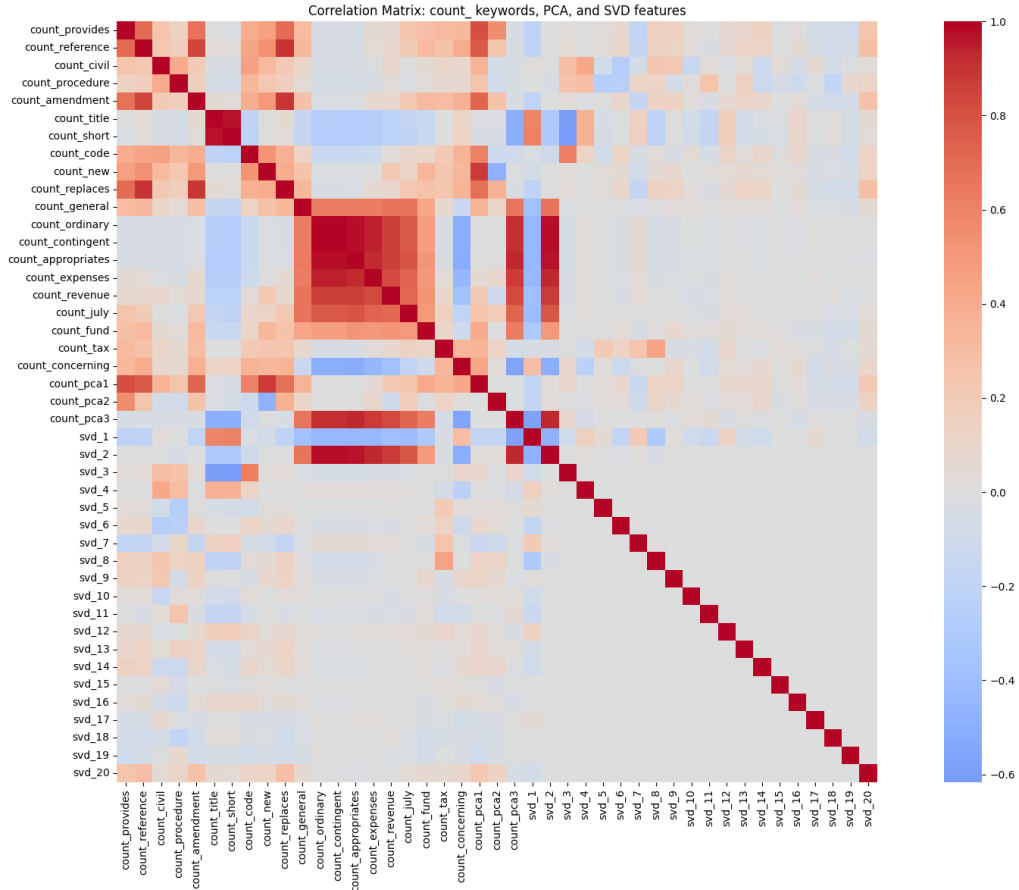


Figure 11: Full correlation matrix between all keyword count features, PCA components, and SVD components. The block of strong correlations among the `count_` variables motivates dimensionality reduction via PCA, while the `svd_` features exhibit low correlation with both count and PCA dimensions. This confirms that SVD-derived semantic topics capture orthogonal information from full bill abstracts, justifying the inclusion of all three text-derived feature sets in modeling.

Appendix B: Confusion Matrices

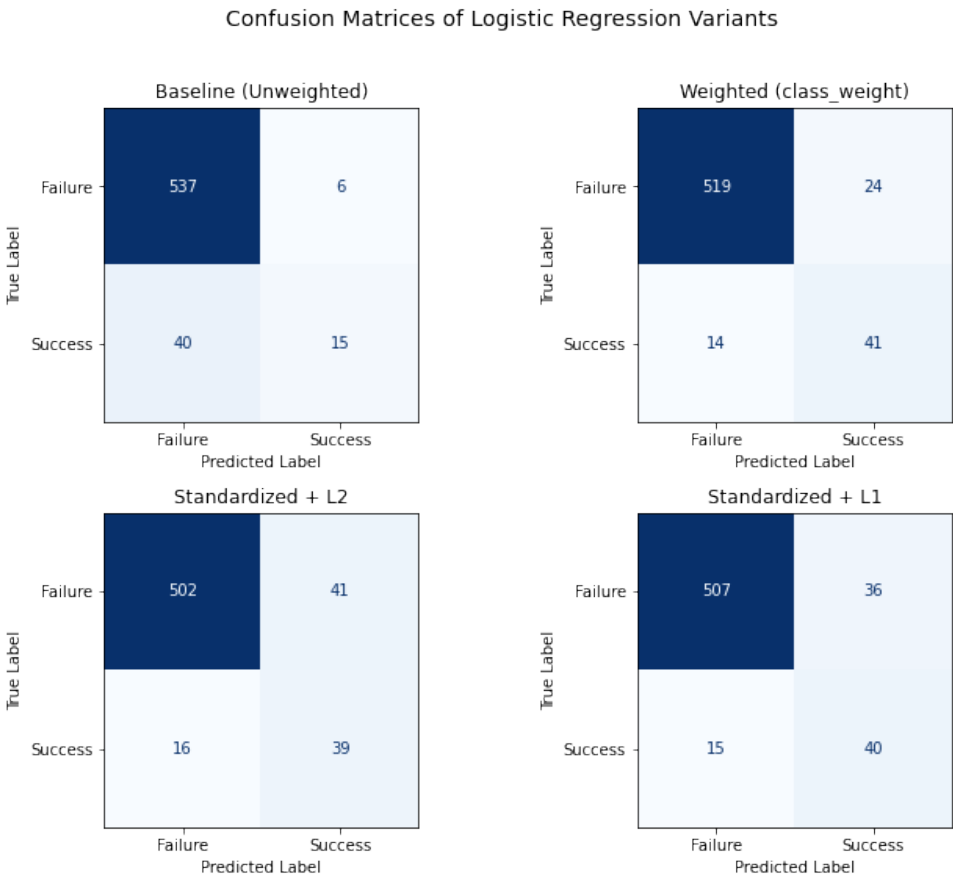


Figure 12: Confusion matrices for four logistic regression variants, evaluated on the same test set. **(Top-left)** The baseline (unweighted) model misclassifies most successful bills as failures, with high precision but poor recall. **(Top-right)** Applying class weights greatly improves recall while preserving overall accuracy. **(Bottom-left)** L2 regularization shows moderate recall gains. **(Bottom-right)** L1 regularization achieves the best balance, identifying more true positives without sacrificing interpretability. All models use the same feature set.

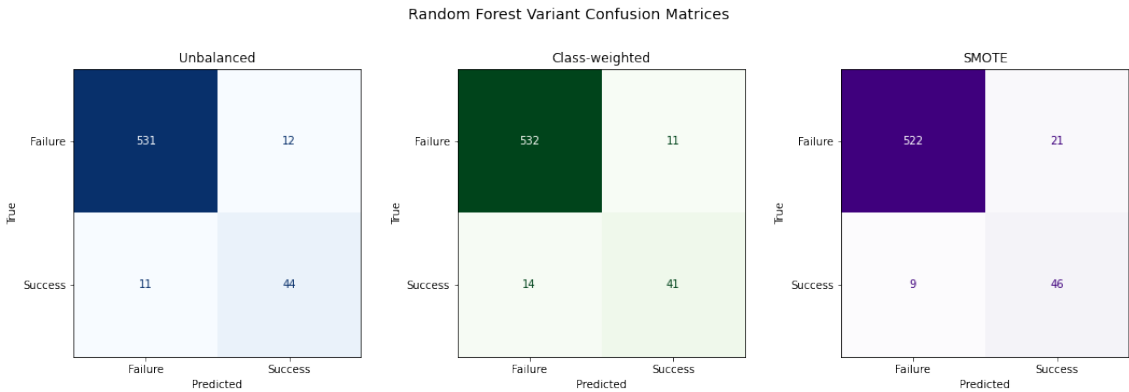


Figure 13: Confusion matrices for three Random Forest variants applied to bill success prediction. **(Left)** The unbalanced model shows strong accuracy but under-identifies the positive class. **(Center)** Reweighting class loss improves recall of successful bills. **(Right)** SMOTE-based oversampling achieves the best recall and balance, visibly reducing false negatives while preserving precision. These visuals support the numeric metrics reported in the main text.

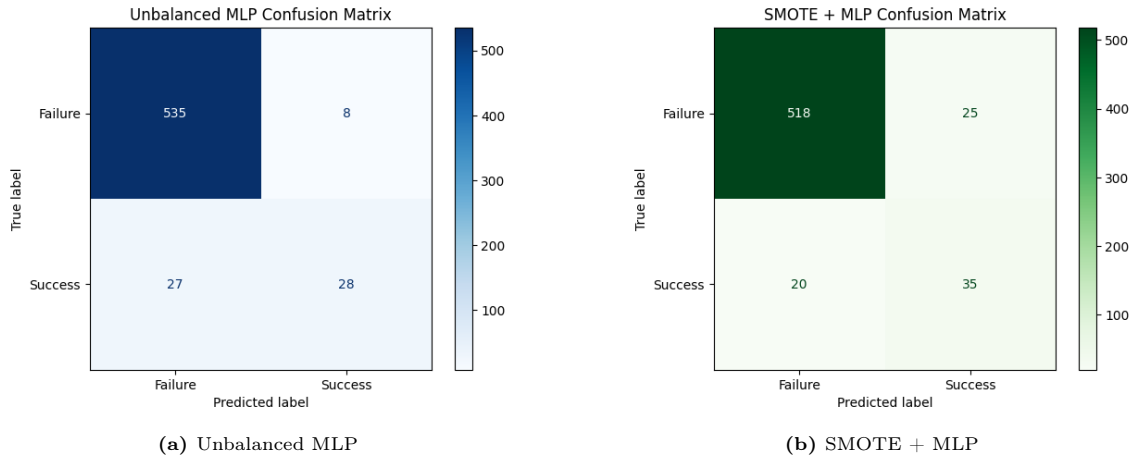


Figure 14: Comparison of confusion matrices for the unbalanced MLP (left) and SMOTE-augmented MLP (right). True Positives (TP) increase from 28 to 35, indicating improved detection of successful bills. False Negatives (FN) decrease from 27 to 20, while False Positives (FP) increase from 8 to 25, reflecting a trade-off in specificity. True Negatives (TN) drop slightly from 535 to 518. Overall, the SMOTE-augmented model improves recall on the minority class (35/55 vs. 28/55), demonstrating the effectiveness of oversampling when prioritizing sensitivity.

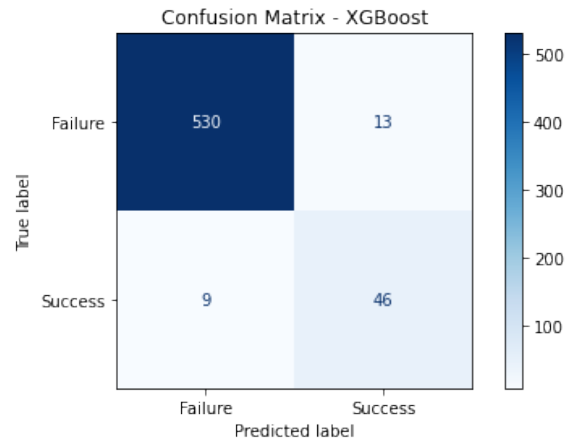


Figure 15: Confusion matrices for XGBoost applied to bill success prediction.

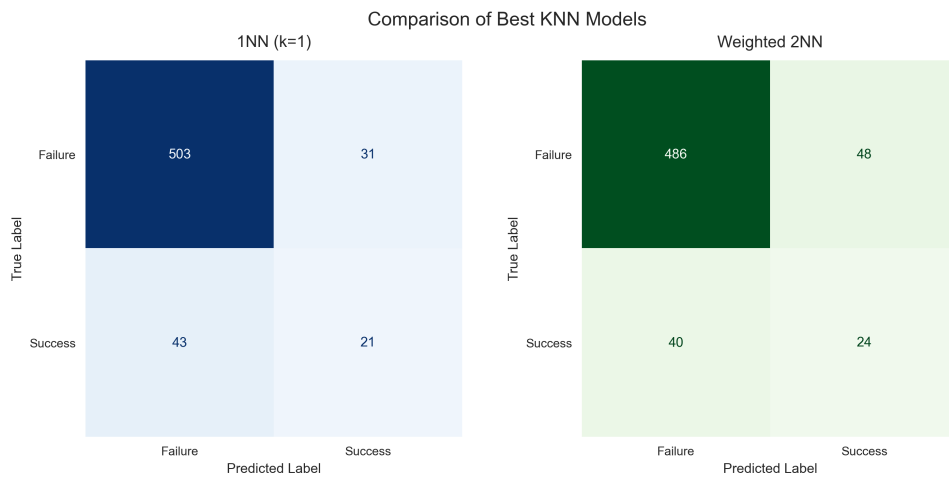


Figure 16: Confusion matrices for KNN applied to bill success prediction. **(Left)** 1NN without cosine similarity weighted. **(Right)** 2NN with cosine similarity weighted.

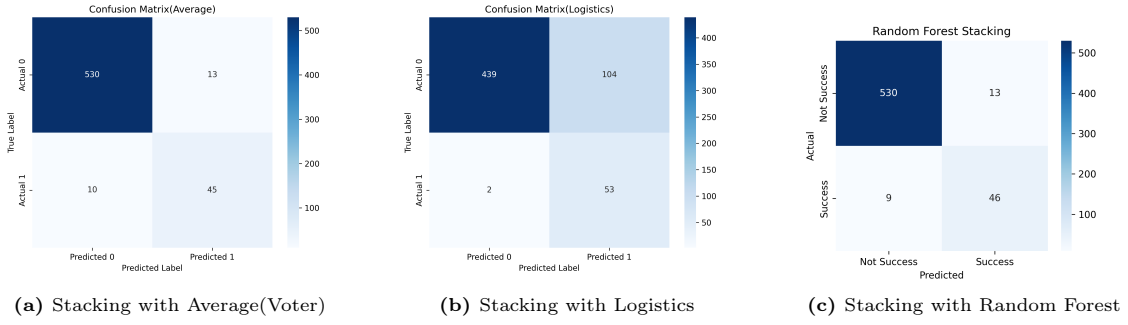


Figure 17: Confusion matrices for Stacking models applied to bill success prediction.

Appendix C: Model Interpretability via SHAP for SMOKE - Random Forest

To provide additional transparency into the model’s behavior, we computed SHAP values to evaluate the marginal contribution of each input feature to the model’s predicted probability of bill success. Figure 18 visualizes these contributions across all test samples for the SMOTE-augmented random forest model.

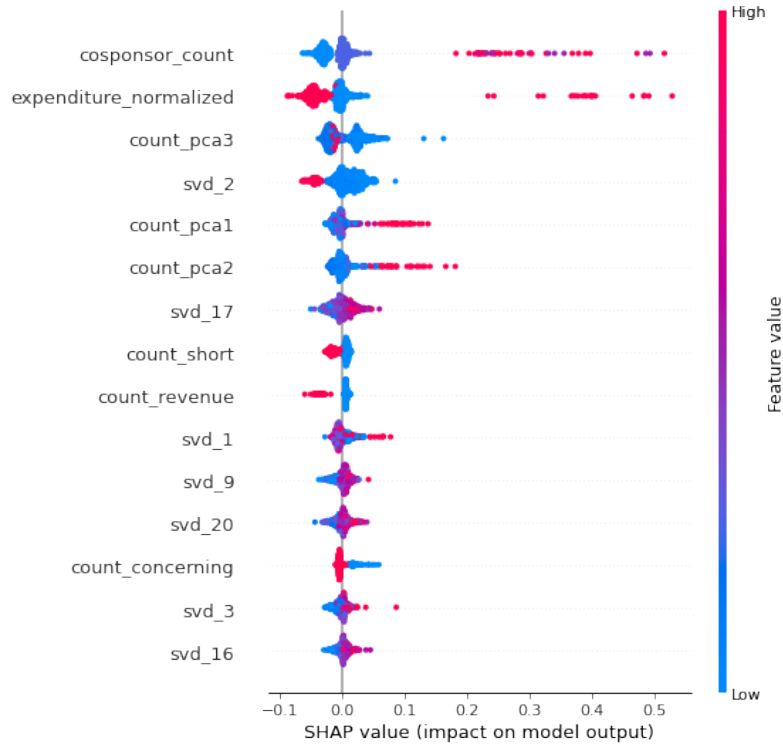


Figure 18: SHAP summary plot of the top 15 predictors from the SMOTE-augmented random forest. Each point represents a feature’s contribution (SHAP value) to a single prediction. Color indicates the feature’s value (red = high, blue = low).

The results confirm the dominance of `cosponsor_count` and `expenditure_normalized`, as well as the contribution of textual components such as `count_pca1`, `svd_2`, and others. The color coding and spread of points also highlight non-linear interactions and heterogeneity in feature effects across samples.

Appendix D: Model Interpretability via SHAP for XGBoost

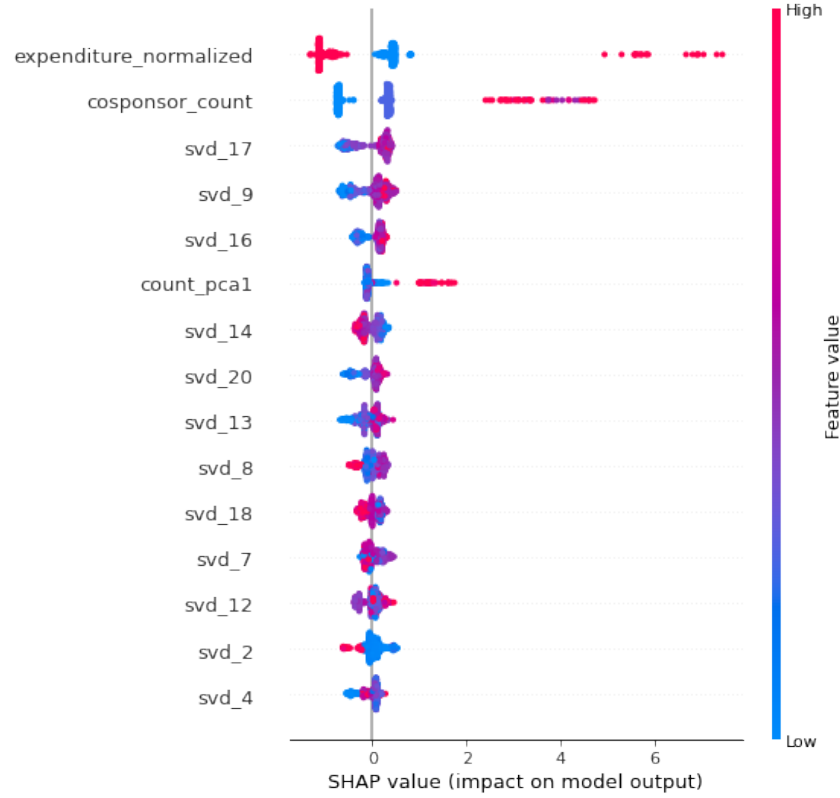


Figure 19: SHAP summary plot of the XGBoost model, illustrating the top 15 features' marginal contributions to the predicted probability of legislative success. Each point represents a bill in the test set, where color encodes the feature value (red = high, blue = low) and horizontal position indicates SHAP value (i.e., impact on model output). Features such as `expenditure_normalized` and `cosponsor_count` exhibit high and consistent positive influence on success predictions. Meanwhile, various `svd_` components also play meaningful roles, capturing latent topics in bill content beyond structured metadata.